~~MEASURING AND ENHANCING TRUSTWORTHINESS OF LLMS IN RAG THROUGH GROUNDED ATTRIBUTIONS AND LEARNING TO REFUSE~~

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01 Title & Authors

- We present this work by **Maojia Song**, **Shang Hong Sim**, **Rishabh Bhardwaj**, **Hai Leong Chieu**, **Navonil Majumder**, and **Soujanya Poria**.
- We represent a collaboration across two institutions.
  - We are affiliated with **SUTD** and **DSO National Laboratories**.
- We invite inquiries via the listed contacts.
  - Student emails use **mymail.sutd.edu.sg**; faculty/DSO contacts are provided.

02 Why RAG Needs Trustworthy, Grounded Outputs

- We focus on fully grounded answers using provided documents.
  - We restrict claims to **attached evidence**.
  - We substantiate each statement with **inline citations**.
- We refuse when evidence is insufficient or absent.
  - We abstain with a **generic refusal**.

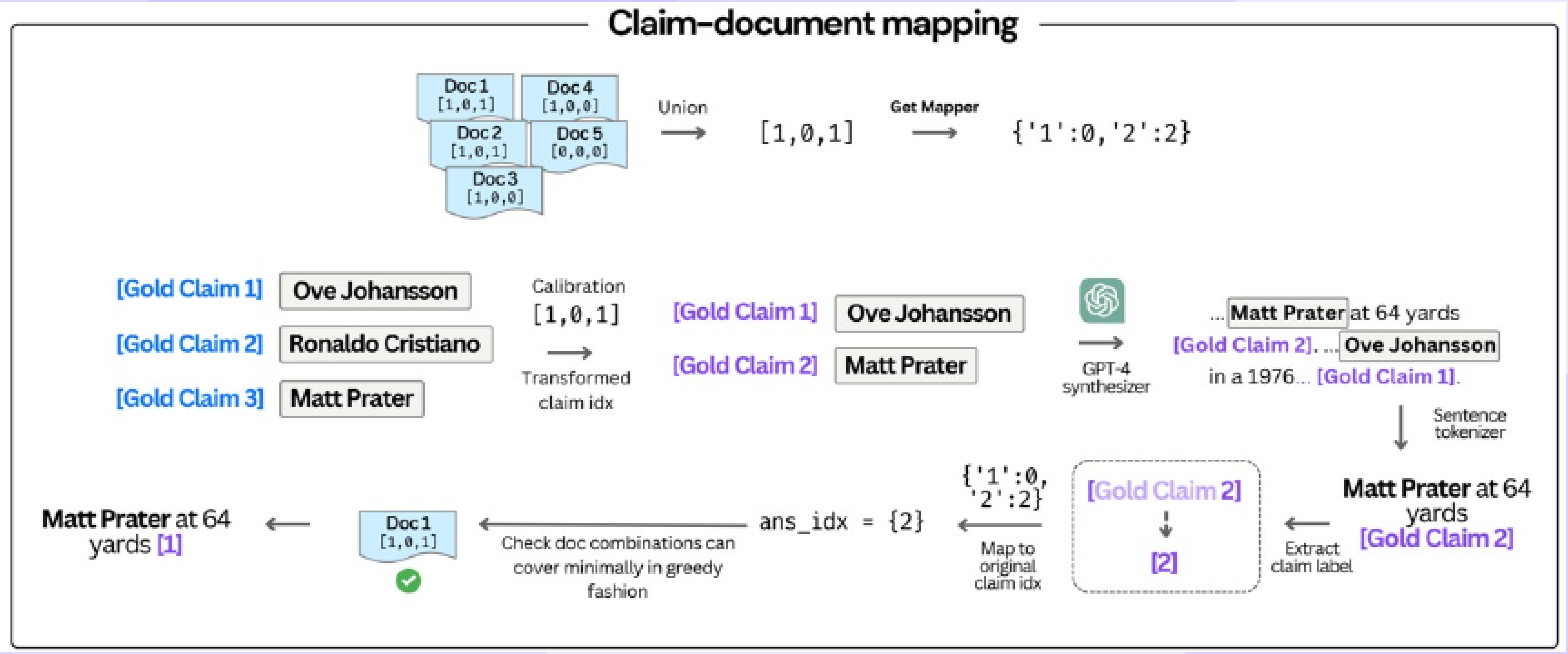
- We evaluate **answer rate**, **correctness**, and **grounding quality**.
  - We assess refusal accuracy and **answer calibration**.
  - We check citation **substantiation and relevance**.

03 TRUST-SCORE: Retriever-agnostic Groundedness Evaluation

- We isolate **model behavior** from retriever quality.
  - We avoid dependence on **document recall**.
  - We enable consistent cross-dataset comparisons.
- We measure **grounded refusals** and **answer correctness**.
  - We score correctness on **answerable cases**.
  - We verify whether refusals are **appropriate**.

- We score citation **supportedness** and **relevance**.
  - We require statements be **entailed by citations**.
  - We avoid irrelevant sources.

04 Task and Outputs: Grounded Statements with Inline Citations

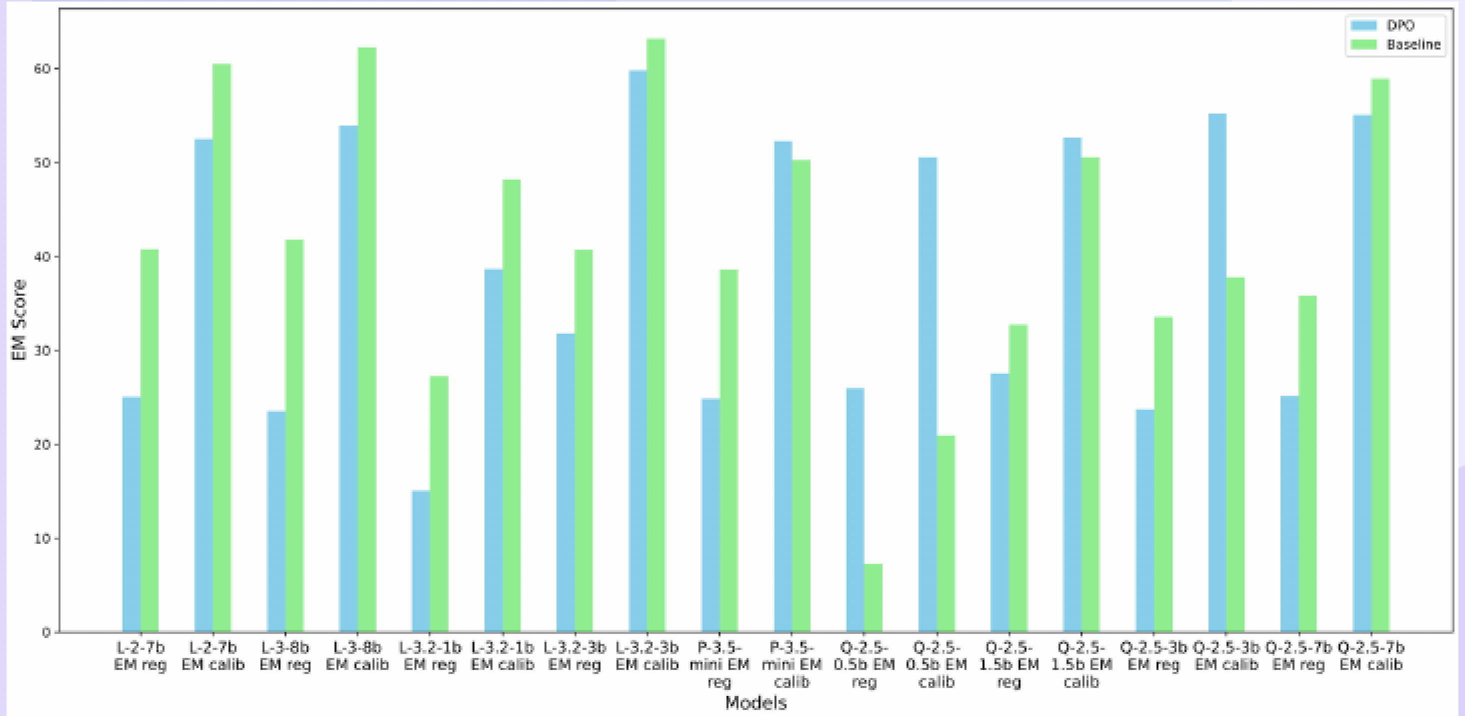


- We take an input **question** with **retrieved documents**.
  - We rely on documents as **sole evidence**.
  - We treat answerability as a function of **document content**.
- We produce statements with **inline citations**.
  - We link each claim to **supporting sources**.
  - We allow distinct claims to cite **distinct documents**.
- We refuse with no claims and no citations when needed.
  - We use a **generic inability** message.
  - We either stay grounded or **explicitly abstain**.

05 Hallucination Taxonomy and Motivation for a New Metric

- We characterize answer errors: inaccurate answers and over-responsiveness.
  - We flag over-responsiveness when **refusal is warranted**.
  - We avoid excessive refusal that hides recoverable answers.
- We characterize citation issues: overcitation and improper citation.
- We motivate a **unified groundedness** metric.
  - We treat complete groundedness as an **IR problem**.
  - We align trustworthiness with **evidence use**.

06 Why Existing Metrics Fail for LLM-in-RAG



- We evaluate truthfulness against **document-obtainable** content.
  - We match generated claims to **available evidence**.
  - We note parametric knowledge can distort scoring.
- We validate attribution via **claim–citation support**.
  - We reject unsupported citations that misrepresent **grounding**.
  - We enforce relevance to avoid off-topic sources.
- We show prior metrics suffer recall consolidation and gamification.
  - We find upper bounds vary with **document coverage**.
  - We observe parametric models inflate **recall unfairly**.

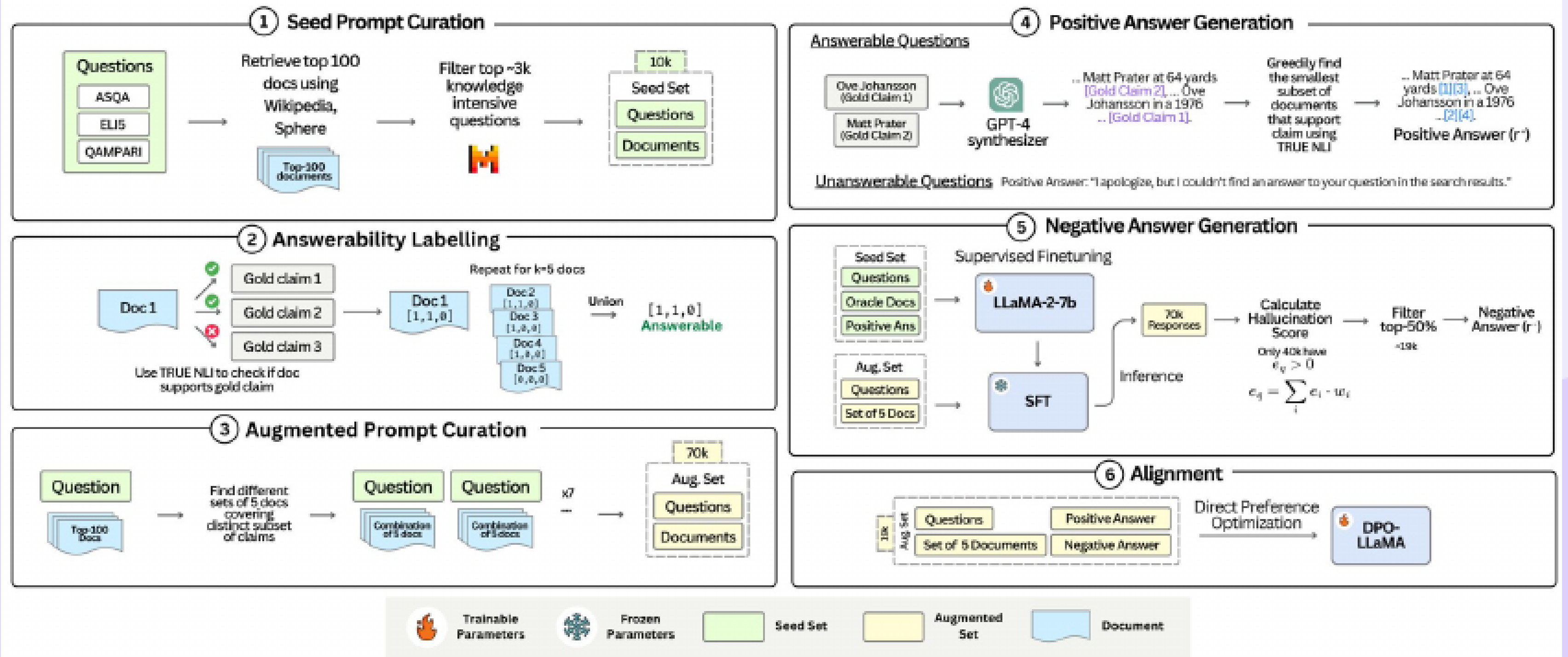
07 Answer Calibration and Grounded Refusals

- We calibrate recall to **obtainable claims**.
  - We make per-sample recall **comparable**.
  - We aggregate dataset scores **fairly**.
- We report **PAC**, **RAC**, and **F1_AC**.
  - We reward **precision among answers** via PAC.
  - We reward **coverage of answerables** via RAC.
- We combine **F1 for refusals** and **F1 for answers**.
  - We reflect **abstention quality**.
  - We promote robust behavior.

08 Citation Groundedness and Final TRUST-SCORE

- We evaluate **support and coverage** for attribution.
  - We compute citation recall for **statement support**.
  - We compute citation precision for **source relevance**.
- We form **F1 for grounding** from precision and recall.
  - We balance precision and recall with a **harmonic mean**.
  - We penalize overcitation and mismatch.
- We define TRUST-SCORE as the average of **refusals**, **answers**, and **citations**.
  - We also report **answering rate** (responsiveness).
  - We target selective answering.

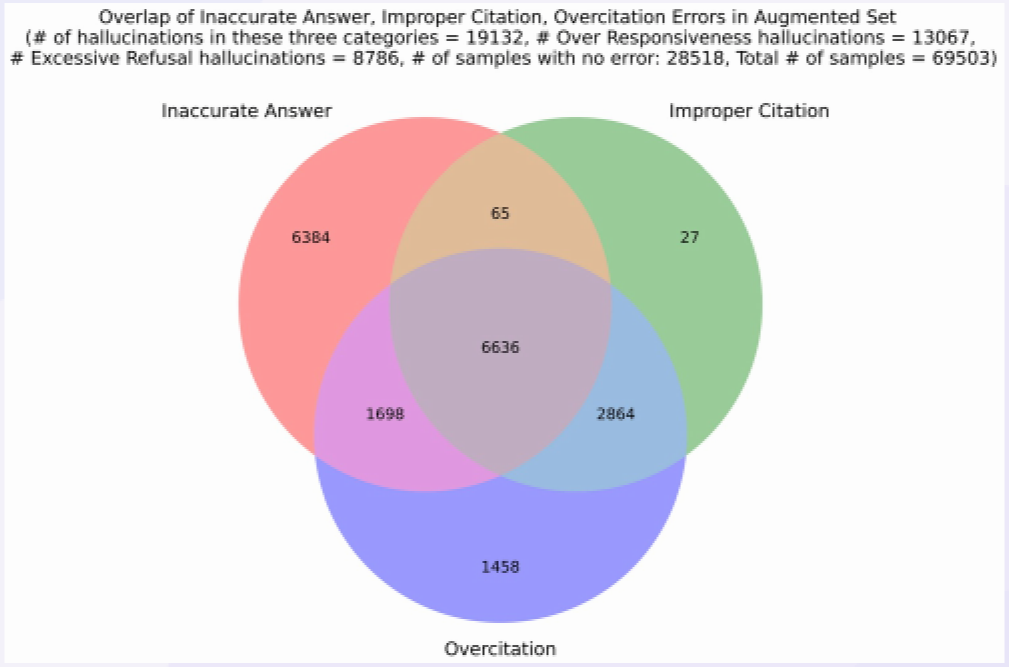
09 TRUST-ALIGN: Dataset and DPO Alignment (**19**K Pairs)



- We construct **preferred** and **unpreferred** responses per prompt.
  - We ensure preferred responses include **citations and coverage**.
  - We elicit unpreferred responses to capture hallucinations.
- We pipeline question curation, retrieval, and **augmentation**.
  - We use gold claims to guide **positives**.
- We scale to 19K pairs from **70K** variants.
  - We apply DPO to align toward **grounded outputs**.
  - We include refusals for **unanswerables**.

10 Targeted Hallucination Coverage and Pair Selection

- We rank negatives by **hallucination severity**.
  - We aggregate **typed errors**.
  - We weight **harmful patterns**.
- We balance **answerable** and **unanswerable** prompts.
  - We sample top cases for effective supervision.
  - We curb bias toward easy errors.
- We emphasize **coverage of failure modes**.
  - We target **refusal and citation** mistakes.
  - We train DPO for **selective, grounded** behavior.



11 Experimental Setup: Models, Datasets, Baselines

- We evaluate **LLaMA**, **Qwen**, and **Phi**.
  - We test vanilla and **TRUST-ALIGN** checkpoints.
  - We span **small to medium** sizes.
- We benchmark on **ASQA**, **QAMPARI**, and **ELI5**.
  - We include **ExpertQA** for OOD.
  - We append top-5 documents per query.
- We compare to **ICL**, **PostCite**, **PostAttr**, **Self-RAG**, and **FRONT**.
  - We attach **post-hoc citations** for closed-book baselines.
  - We target **attribution quality** in Self-RAG and FRONT.

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Results at a Glance: **26/27** TRUST-SCORE Improvements

- We deliver consistent TRUST gains in 26/27 settings.
- We show large jumps on **LLaMA-3-8B** across datasets.
- We note mixed F1_AC under stricter refusals.

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Understanding Mixed Effects on Answer Correctness

- We raise PAC via selective answering on hard items.
  - We remove **borderline errors**.
  - We improve **precision** among answers.
- We lower RAC when answerable items are refused.
  - We reduce **coverage** of answerables.
  - We keep the denominator fixed while **answers shrink**.
- We still improve TRUST through **better refusals** and **cleaner citations**.
  - We raise the **composite score**.
  - We reflect **calibrated behavior**.

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Qualitative Examples: Refusals and Citations (2-up)

- We refuse without claims when **evidence is missing**.
  - We treat unsupported designer names as hallucinations.
  - We avoid confident answers without sources.
- We abstain when **timeframes are ambiguous**.
  - We note pilots/enforcement do not imply legalization.
  - We demand explicit evidence for dates.
- We avoid **precise unsupported values** on near-miss evidence.
  - We distinguish captive vs. wild lifespan.
  - We prevent fabricated numbers.

Qualitative Examples: Citation Support vs Improper/Over-Citation

- We prefer minimal sufficient citations for **precision and clarity**.
  - We cite a single authoritative source when possible.
  - We avoid redundant sources.
- We reject **improper citations**.
  - We do not infer years from specs without dates.
  - We refuse when support is absent.
- We separate multi-claim answers with **distinct sources**.
  - We map founders and year to relevant docs.
  - We avoid bundling that leaves claims unsupported.

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Qualitative Examples: PAC vs RAC in Practice

- We increase **precision among answers** via selective refusals.
  - We raise PAC by filtering **borderline items**.
  - We reduce incorrect answers.
- We lower **recall over answerables** when we abstain.
  - We see RAC decline on partially answerable items.
  - We embrace calibrated caution.
- We improve composite trust via **refusals and citations**.
  - We increase F1 for grounded refusals.
  - We strengthen F1 for citation grounding.
- We see these behaviors reflected in higher F1_GR and F1_GC.

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Qualitative Vignette: ASQA Ambiguity

- We pose: When did **Aurora** first launch?
- We observe D snippets conflict (software vs festival).
  - Doc A: "Aurora (software) entered public beta in 2019."
  - Doc B: "The Aurora cultural festival began in 2018."
- We see baseline answer: "Aurora launched in 2019 [A]."
- We, TRUST-ALIGN, refuse: "Entities differ; we abstain."

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Qualitative Vignette: QAMPARI Incomplete List

- We ask: Name four countries that border the Black Sea.
- We find D mentions at most three.
  - Doc 1: "Black Sea borders Turkey and Bulgaria."
  - Doc 2: "Romania has a coastline on the Black Sea."
- We see baseline: adds **Ukraine** with shaky cites.
- We, TRUST-ALIGN, refuse: "Docs support ≤3; we abstain."

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Qualitative Vignette: Partial Answerability (PAC vs RAC)

- We ask: Who co-founded Project Aurora and in which year?
- We read D: year (2014) + one co-founder (Alice Nguyen).
- We see baseline answer: adds **Bob Kim** and wrong year.
- We, TRUST-ALIGN, refuse: "All co-founders not supported; abstain."

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Case Study: ASQA vs QAMPARI contrasts

- We see ASQA benefit from **selective answering** and precise citations.
  - We improve grounded refusals and consistency.
  - We accept mixed changes in answer correctness.
- We see QAMPARI gain from **calibrated list answers**.
  - We tie distinct items to **distinct sources**.
  - We reduce spurious additions.
- We improve attribution by aligning **items and sources**.
  - We keep evidence ties traceable.
  - We decrease overcitation.

20 Case Study: ELI5 decomposition trade-offs

21 Generalization and Objective: Families, Sizes, and DPO vs SFT

- We generalize across **model families and sizes**.
- We find DPO outperforms SFT on **TRUST and citations**.
- We demonstrate **scalable alignment** for RAG.

22 Ablations, OOD Generalization, and Parametric Knowledge Use

- We show refusal data is **crucial for grounding**.
- We generalize out of domain better than baselines.
- We reduce parametric reliance via alignment.

23 Main Failure Mode and Weaknesses

- We observe NLI sensitivity introducing label noise.
  - We can misestimate answerability and grounding.
  - We can propagate noisy training signals.
- We note risks of metric gamification and granularity drift.
  - We see parametric facts paired with unsupported cites.
  - We find decomposition mismatches skew scoring.

- We are mitigating via **ensembles**, **adjudication**, and **audits**.
  - We run adversarial tests to expose weaknesses.
  - We tune granularity to improve coverage.
- We found NLI misses paraphrased support, leading to **false refusals**.

Conclusion, Practical Guidance, and Limitations

- We offer **holistic groundedness** with TRUST-SCORE for RAG.
  - We score **answers**, **refusals**, and **citations**.
  - We control for retriever effects.
- We reduce hallucinations with TRUST-ALIGN + DPO.
  - We bring open-weight models toward **strong performance**.
- We recommend **refusal data**, **DPO**, and **citation audits**.
  - We monitor answering rate for over-responsiveness.
  - We prioritize **precision over coverage** under uncertainty.

Thank You

Questions & Discussion